

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – II
PAPER –2
TEST DATE: 10-12-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D
 Sol. During motion of ball its magnetic flux linked with coil will be zero at any time and therefore no induced current will develop in ring

2. A
 Sol. Potential difference between inner and outer surface of capacitor is same every where, therefore net charge density at each point of inner surface will be same. Hence, there is induced charge on inner surface on the portion which touches the medium and board charge is of opposite nature than that of free charge

$$\Rightarrow (\sigma_{\text{free}} - \sigma_{\text{induced}})_{\text{at } x} = (\sigma_{\text{free}})_{\text{at } y}$$

$$\Rightarrow \sigma_{\text{free}} \text{ at point } X > \sigma_{\text{free}} \text{ at point } Y$$

Let $\sigma_f \rightarrow$ charge density of free charge at X

$\sigma_i \rightarrow$ charge density of induced charge near X

$\sigma'_f \rightarrow$ charge density of free charge at Y

$$(\sigma_f + \sigma_i) = \frac{\sigma_f}{K} = \sigma'_f$$

$$\sigma_i = \sigma_f \left(1 - \frac{1}{K} \right) \quad \dots(i)$$

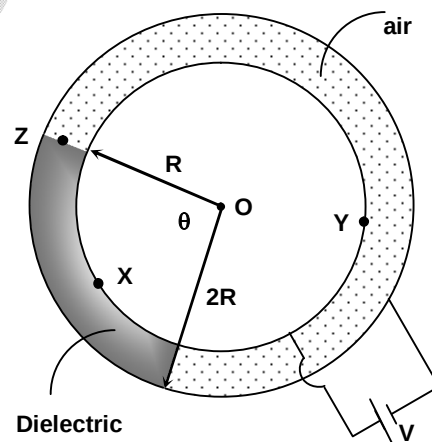
$$\text{Also, } K\sigma'_f = \sigma_f$$

$$\sigma_f + \sigma_i = \sigma'_f$$

$$\sigma'_f + \sigma_i = \sigma'_f$$

$$\sigma'_f = \frac{\sigma_i}{1-K} = -\frac{\sigma_i}{K-1}$$

$$\text{In air gap, } \frac{\sigma'_f 4\pi R^2}{4\pi\epsilon_0 r^2} dr = dV$$



$$\int_R^{2R} \frac{\sigma_f' R^2}{\epsilon_0} \cdot \frac{dr}{r^2} = \int dV$$

$$V = \frac{\sigma_f' R^2}{\epsilon_0} \left(\frac{1}{2R} \right) \text{ or } \sigma_f' = \left(\frac{2\epsilon_0 V}{R} \right)$$

3. D
Sol. Basic concept

4. B
Sol. $J = \frac{3}{\pi(2)^2 - \pi(1)^2} = \frac{1}{\pi} \text{ A/m}^2$

There is uniform magnetic field inside cavity

$$B = \frac{\mu_0 J r}{2} = \frac{4\pi \times 10^{-7} \times \frac{1}{\pi} \times 1}{2} = 0.2 \times 10^{-6} \text{ T}$$

Let particle collides at point A

$$\Rightarrow x = (v_0 \sin \theta) \frac{T}{4}$$

$$= v_0 \sin \theta \times \frac{1}{4} \times \frac{2\pi m}{qB} = \left[\frac{3\pi m v_0 \sin \theta}{qB} \right] = \left(\frac{\pi v_0 \sin \theta}{2} \right)$$

For no collision

$$\left(\frac{R}{2} \right)^2 > r^2 + x^2$$

$$1 > (v_0 \cos \theta)^2 + \left(\frac{\pi v_0 \sin \theta}{2} \right)^2 = v_0^2 \left(\cos^2 \theta + \frac{\pi^2}{4} \sin^2 \theta \right) = v_0^2 \left[1 + \left(\frac{\pi^2}{4} - 1 \right) \sin^2 \theta \right]$$

$$v_0 < \frac{1}{\sqrt{1 + \left(\frac{\pi^2}{4} - 1 \right) \sin^2 \theta}}$$

$$v_0 < \frac{1}{\sqrt{2}} \text{ m/s}$$

For collision at, $t = \frac{3T}{4}$, $v_0 < \sqrt{\frac{3}{46}} \text{ m/s}$

5. A, C, D

Sol. Time taken by sphere to achieve temperature 40° , $\frac{40 - 20}{2} = 10 \text{ sec}$

Heat absorbed by sphere

$$Q_1 = ms\Delta T$$

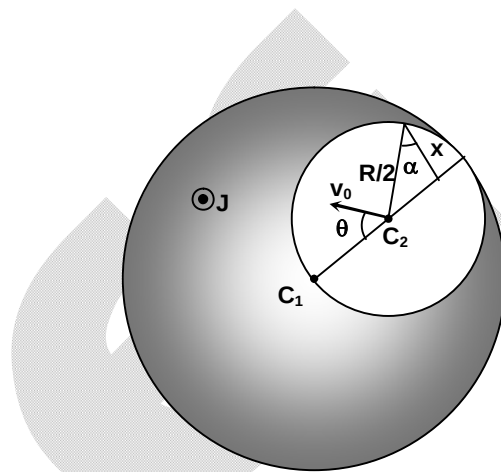
$$= 10^3 \times 0.5 \times 20 = 10^4 \text{ cal}$$

Rate of heat supplied by heater

$$\frac{dQ}{dt} = \left(ms \frac{dT}{dt} \right) + \sigma A (T^4 - T_0^4)$$

Hence, T is changing with time

$\Rightarrow \frac{dQ}{dt}$ is not constant as T is increasing and ΔT is also increasing.



Surface area of sphere is not known, therefore $\frac{dQ}{dt}$ can not be determined

6. B, C

Sol. Let charge on outer surface q is q potential at C_1 , $v_{C_1} = \frac{9 \times 10^7 \cdot q_0}{13} + \frac{9 \times 10^9 \times q'}{4} = 0$

$$\Rightarrow q' = -\frac{4}{13}q_0$$

Potential due to q' at C_1

$$V = -\frac{k \times \frac{4}{13}q_0}{4} = -\frac{kq_0}{13}$$

7. A, B, D

Sol. Given $V_1 = 3\sin(100t + \pi/3)$

$$V_2 = 3\sqrt{3}\sin(100t - \pi/6)$$

$\Rightarrow$ net source voltage $V = 6\sin(100t)$

$$\therefore z = \frac{V_0}{i_0} \Rightarrow \text{for } i = 2\cos(100t)$$

$$Z = 6/2 = 3\Omega$$

If $i = i_0\cos(100t)$, then circuit is non resistance

for $i = i_0\sin(100t)$ circuit is pure resistive

SECTION – B

8. 6

Sol. Let excess pressure exerted by liquid on walls of is P when it is heated from 40°C to 76°C .
During Heating

$$\text{For shell, } |\Delta V_1| = \frac{PV}{B_S}$$

$$\text{For liquid, } |\Delta V_2| = \gamma V \Delta T - \frac{PV}{B_L}$$

$$\therefore |\Delta V_1| = |\Delta V_2|$$

$$\frac{PV}{B_S} = \gamma V \Delta T - \frac{PV}{B_L}$$

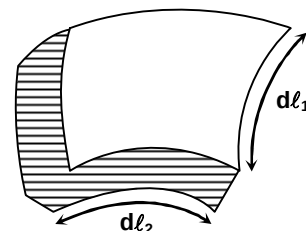
$$\Rightarrow P \left(\frac{1}{B_L} + \frac{1}{B_S} \right) = \gamma \Delta T$$

$$P \left(\frac{2}{B_0} + \frac{1}{B_0} \right) = \gamma \Delta T$$

$$\Rightarrow P = \frac{\gamma B_0 \Delta T}{3}$$

$$Pd\ell_1 \cdot d\ell_2 = (S \cdot d \cdot d\ell_1) \frac{d\ell_2}{R} + (S \cdot d \cdot d\ell_2) \frac{d\ell_1}{R}, \quad \frac{\gamma B_0 \Delta T}{3} = \frac{2S \cdot d}{R}$$

$$S = \frac{36\gamma B_0 R}{6d} = \frac{6\gamma B_0 R}{d}$$



9. 3

 Sol. When S_1 open and S_2 is closed

$$i_{\max} = \left(\frac{E_0}{R_0} \right)$$

 Now S_2 is open, S_1 is closed (just after)

$$i = \frac{E_0}{3R_0} \text{ and charge on capacitor is zero}$$

Maximum charge on capacitor

$$\frac{1}{2} \frac{q_0^2}{C_0} = \frac{1}{2} (3L_0) \frac{E_0^2}{9R_0^2}$$

$$q_0 = \sqrt{\frac{L_0 C_0 E_0^2}{3R_0^2}} = \frac{E_0}{R_0} \sqrt{\frac{L_0 C_0}{3}}$$

 Charge on capacitor at time t

$$q = q_0 \sin(\omega_0 t), \text{ where } \omega_0 = \frac{1}{\sqrt{3L_0 C_0}}$$

Energy stored in capacitor

$$U = \frac{q^2}{2C}$$

$$\frac{dU}{dt} = \frac{1}{2C} \cdot 2q \cdot \frac{dq}{dt} = \frac{1}{C} \cdot q_0 \sin \omega_0 t \cdot q_0 \omega_0 \cos \omega_0 t$$

$$\frac{dU}{dt} = \frac{q_0^2 \omega_0}{2C} \sin 2\omega_0 t$$

 For $\frac{dU}{dt}$ to be maximum $\sin 2\omega_0 t = 1$

$$\Rightarrow \omega_0 t = \pi/4$$

$$\Rightarrow i = \frac{d}{dt} q_0 \sin \omega_0 t$$

$$= q_0 \omega_0 \cos \omega_0 t = \left(\frac{q_0 \omega_0}{\sqrt{2}} \right)$$

$$= \frac{E_0}{\sqrt{2}R_0} \sqrt{\frac{L_0 C_0}{3}} \cdot \frac{1}{\sqrt{3L_0 C_0}} = \frac{E_0}{3\sqrt{2}R_0} = \frac{3}{\sqrt{2}} A$$

10. 8

 Sol. $\tau = MB \sin \theta = iAB \sin \theta = iAB \sin \theta$

$$= \frac{1}{R} \frac{d\phi}{dt} AB \sin \theta$$

$$\Rightarrow \tau dt = \frac{1}{R} (BA \sin \theta \cdot d\theta) AB \sin \theta = \frac{A^2 B^2}{R} \sin^2 \theta d\theta$$

$$\int \tau dt = \frac{A^2 B^2}{2R} \int_0^\pi (1 - \cos 2\theta) d\theta = \frac{\pi A^2 B^2}{2R}$$

$$\text{Average torque} = \frac{\int \tau dt}{t} = \frac{A^2 B^2}{2R}$$

$$B = \frac{\mu_0 i}{2\pi a}$$

$$\text{Average torque} = \frac{b^2}{2R} \left(\frac{\mu_0 i}{2\pi a} \right)^2 = \frac{\mu_0^2 i^2 b^2}{8\pi^2 a^2 R}$$

11. 3

Sol. $W_1 = 0$ as configuration of the field is same in both the cases

$$W_2 = -\frac{1}{2} \epsilon_0 \int_{R/4}^R \left(\frac{q_0}{4\pi \epsilon_0 r^2} \right)^2 4\pi r^2 dr = -\frac{3q_0^2}{8\pi \epsilon_0 R}$$

12. 7

Sol. Area of ring = $\pi r^2 = 1\text{m}^2$ Torque of ring = $iAB \sin \theta = 1 \times 1 \times 2 \sin \theta (90^\circ + 37^\circ)$

$$20 \times \frac{4}{5} = 16 \text{ N-m (clockwise)}$$

$$N_2 = f = \mu N_1$$

$$N_1 = 20 \text{ N}$$

$$\Rightarrow N_2 = 20 \mu$$

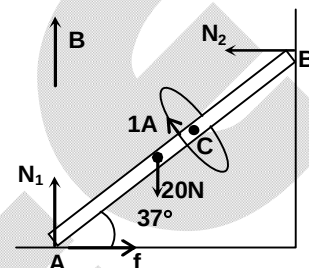
Taking torque about A

$$N_2 \times 5 \sin 37^\circ - 20 \times \frac{5}{2} \cos 37^\circ - 16 = 0$$

$$20\mu \times 3 - 40 - 16 = 0$$

$$\mu = \frac{14}{15}$$

... (i)
... (ii)



13. 2

Sol. When S is open

$$\text{Let } V_{P_3} = V_{Q_3} = 0$$

$$\Rightarrow V_{P_2} = V_{Q_2} = \frac{2V}{5}$$

$$V_{P_1} = V_{Q_1} = \frac{3V}{5}$$

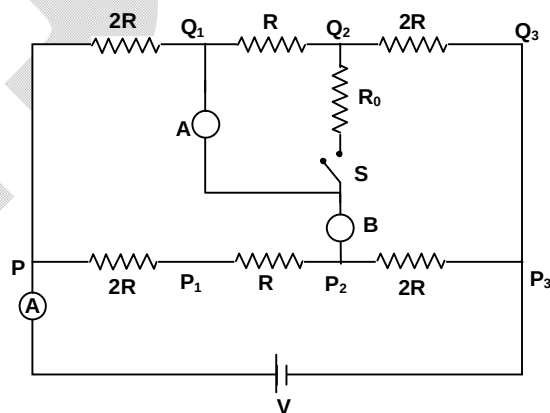
Let current through voltmeter is i (small current) Resistance of voltmeter B is R_0 $\Rightarrow$ Resistance of voltmeter A is $2R_0$

$$\Rightarrow V_{Q_3} - 3R_0 i = V_{P_2}$$

$$\frac{3V}{5} - 3R_0 i = \frac{2V}{5}$$

$$\Rightarrow 3R_0 i = \frac{V}{5}$$

$$R_0 i = V/15$$

Reading of voltmeter A = $2V/15$ Reading of voltmeter B = $V/15$ 

When S is closed

$$V_C = 2V/5$$

For A

$$V_{Q_1} - 2R_0 i_1 - R_0 \frac{i_1}{2} = V_{P_2}$$

$$\frac{3V}{5} - \frac{5}{2} R_0 i_1 = \frac{2V}{5}$$

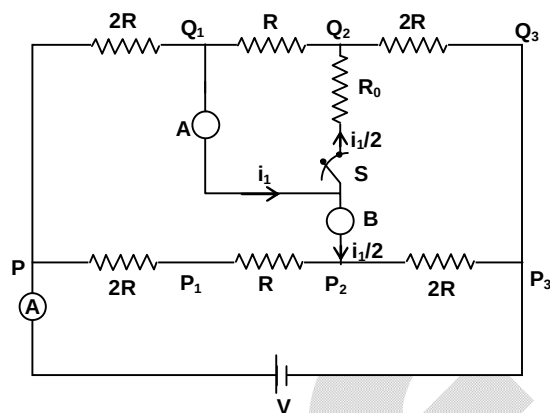
$$\frac{5}{2} R_0 i_1 = V/5$$

$$R_0 i_1 = 2V/25$$

$$\text{Reading of voltmeter A is } \frac{4V}{25}$$

Change in reading of voltmeter A

$$= \frac{2V}{15} - \frac{4V}{25} = \frac{2V}{75}$$



SECTION - C

14. 0.49

Range (0.47 to 0.51)

Sol. $CdT = PdV + C_v dT$

$$C = C_v + P \frac{dV}{dT} \quad \dots(i)$$

for process B $\rightarrow$ C

$$P - 2P_0 = \frac{P_0 - 2P_0}{2V_0 - V_0} (V - V_0)$$

$$P - 2P_0 = -\frac{P}{V_0} (V - V_0) = -\frac{P_0}{V_0} V + P_0$$

$$P = -\frac{P_0}{V_0} V + 3P_0 \quad \dots(ii)$$

$$\frac{RT}{V} = -\frac{P_0}{V_0} V + 3P_0$$

$$RT = -\frac{P_0}{V_0} V^2 + 3P_0 V$$

Differentiating w.r.t T.

$$R = \left(-\frac{P_0}{V_0} 2V + 3P_0 \right) \frac{dV}{dT} \Rightarrow \frac{dV}{dT} = \frac{R}{3P_0 - \frac{2P_0}{V_0} V} \quad \dots(iii)$$

$$\text{Putting } \frac{dV}{dT} \text{ in equation (i), } C = \frac{3}{2} R + \frac{RP}{\left(3P_0 - \frac{2P_0}{V_0} V \right)}$$

For $dQ > 0$

$$\left[\frac{3}{2} R + \frac{RP}{3P_0 - \frac{2P_0}{V_0} V} \right] dT > 0$$

$$\text{Or } 9P_0 - \frac{6P_0}{V_0} V + 2P > 0$$

$$9P_0 - \frac{6P_0}{V_0} + 6P_0 - \frac{2P_0}{V_0} V > 0$$

$$15P_0 - \frac{8P_0}{V_0} V \text{ or } V < \frac{15}{8} V_0$$

Using (i), for $V = \frac{15}{8} V_0$,

$$P = -\frac{P_0}{V_0} \cdot \frac{15}{8} V_0 + 3P_0 = \frac{9P_0}{8}$$

$$T = \frac{\frac{9P_0}{8} \cdot \left(\frac{15}{8} V_0\right)}{R} = \frac{135}{64} \frac{P_0 V_0}{R}$$

Heat absorbed by gas

$$Q = Q_{BD} + Q_{AB}$$

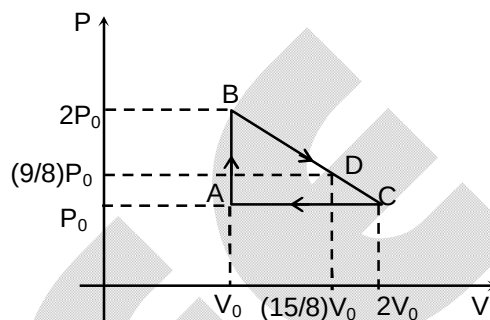
$$Q = W + \Delta U$$

$$= \frac{1}{2} \left[2P_0 + \frac{9}{8} P_0 \right] \left[\frac{15}{8} V_0 - V_0 \right] + \frac{3}{2} R \left[\frac{135}{64} \frac{P_0 V_0}{R} - \frac{2P_0 V_0}{R} \right] + \frac{3}{2} R \left[\frac{2P_0 V_0}{R} - \frac{P_0 V_0}{R} \right]$$

$$= \frac{97}{32} P_0 V_0$$

$$\text{Total work done, } W = \frac{1}{2} (3P_0)(V_0) = \frac{3}{2} P_0 V_0$$

$$\text{Efficiency} = \frac{W}{Q} = \frac{\frac{3}{2} P_0 V_0}{\frac{97}{32} P_0 V_0} = 0.49$$



15. 6.00
Range (5.90 to 6.10)

Sol. $T_C = \frac{P_0 V_0}{R}$ and $T_B = \frac{2P_0 V_0}{R}$

$$\Rightarrow C \rightarrow A \rightarrow B$$

$$W = -\frac{P_0 V_0}{2}$$

$$\Delta U = \frac{3}{2} R \left[\frac{2P_0 V_0}{R} - \frac{P_0 V_0}{R} \right] = \frac{3}{2} P_0 V_0$$

$$Q = \frac{P_0 V_0}{2} + \frac{3}{2} P_0 V_0 = P_0 V_0$$

$$= 6 \times 10^5 \text{ J}$$

16. 4.00
(3.98 to 4.02)

17. 0.707
(0.68 to 0.72)

Sol. $\frac{dv}{dt} = -\frac{b_0 v}{m}$

$$\int_{v_0}^v \frac{dv}{v} = -\frac{b_0}{m} \int_0^t dt$$

$$v = v_0 e^{-\frac{b_0}{m}t}$$

$$\int ds = v_0 \int e^{-\frac{b_0}{m}t} dt$$

$$= \frac{mv_0}{b_0} \left(1 - e^{-\frac{b_0}{m}t} \right) = \frac{m}{b_0} (v_0 - v)$$

$$s \propto m$$

$$\propto (v_0 - v)$$

$$\Rightarrow s = 4 \text{ cm}$$

Motion along x-axis

$$-m_0 \frac{dv_x}{dt} = q_0 v B \sin \theta + b_0 v \cos \theta$$

$$-m_0 \frac{dv_x}{dt} = q_0 B v_y + b_0 v_x$$

$$\text{Or } -m_0 \int_{v_0}^{v_x} dv_x = q_0 B \int_0^y dy + b_0 \int_0^x dx$$

$$-m_0 (v_x - v_0) = q_0 B y + b_0 x \quad \dots(iv)$$

Motion along y-axis

$$m_0 \frac{dv_y}{dt} = q_0 v B \cos \theta - b_0 v \sin \theta$$

$$m_0 \frac{dv_y}{dt} = q_0 B v_x - b_0 v_y$$

$$m \int dv_y = q_0 B \int dx - b_0 \int dy \quad \dots(iv)$$

$$\Rightarrow m_0 v_y = q_0 B x - b_0 y$$

Add square of equation (iv) and (v)

$$m_0^2 [v_0 - v_x]^2 + m_0^2 v_y^2 = q_0^2 B^2 (x^2 + y^2) + b_0^2 (x^2 + y^2)$$

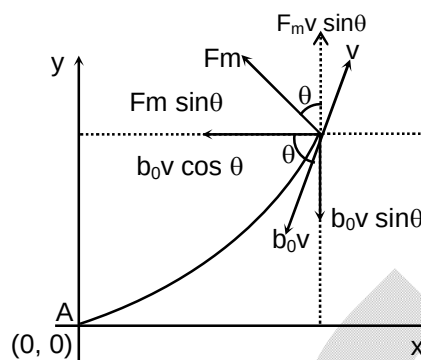
$$m_0^2 (v_0^2 + v_0^2 - 2v_0 v_x) = q_0^2 B^2 r^2 + b_0^2 r^2$$

$$2m_0^2 v_0 (v_0 - v_x) = (q_0^2 B^2 + b_0^2) r^2$$

For $v_x = 0$

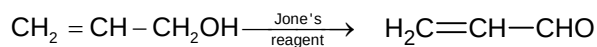
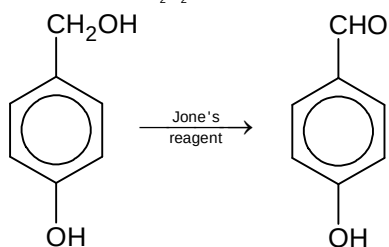
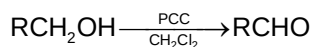
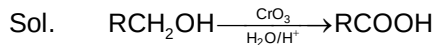
$$2m_0^2 v_0^2 = (q_0^2 B^2 + b_0^2) r^2$$

$$\Rightarrow r = \frac{\sqrt{2} m_0 v_0}{\sqrt{q_0^2 B^2 + b_0^2}} = \frac{\sqrt{2} \times 10^{-3} \times \sqrt{2}}{\sqrt{4 \times 10^{-6} + 4 \times 10^{-6}}} = 0.707 \text{ m}$$

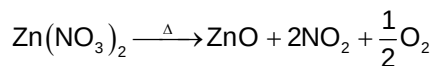
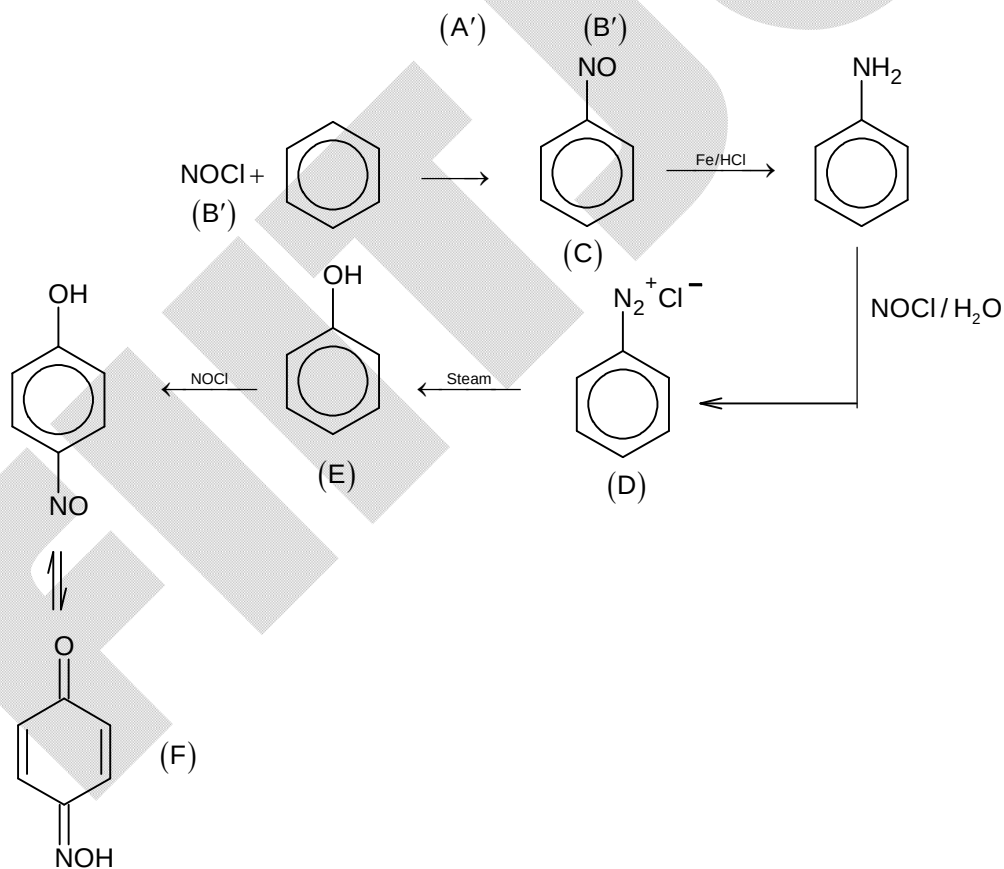
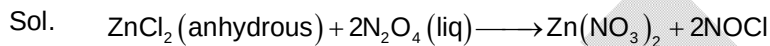


Chemistry**PART – II****SECTION – A**

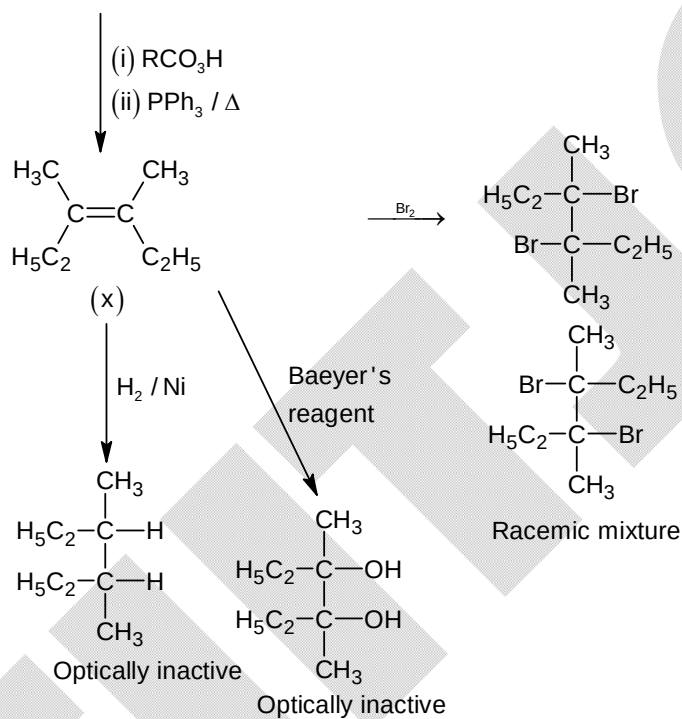
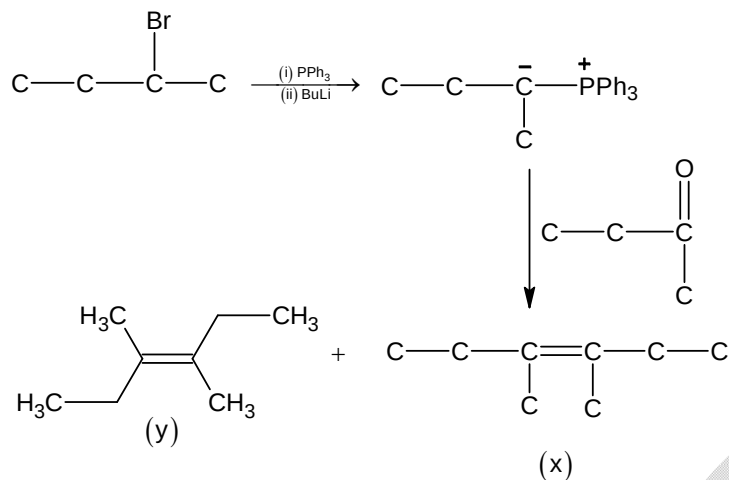
18. C



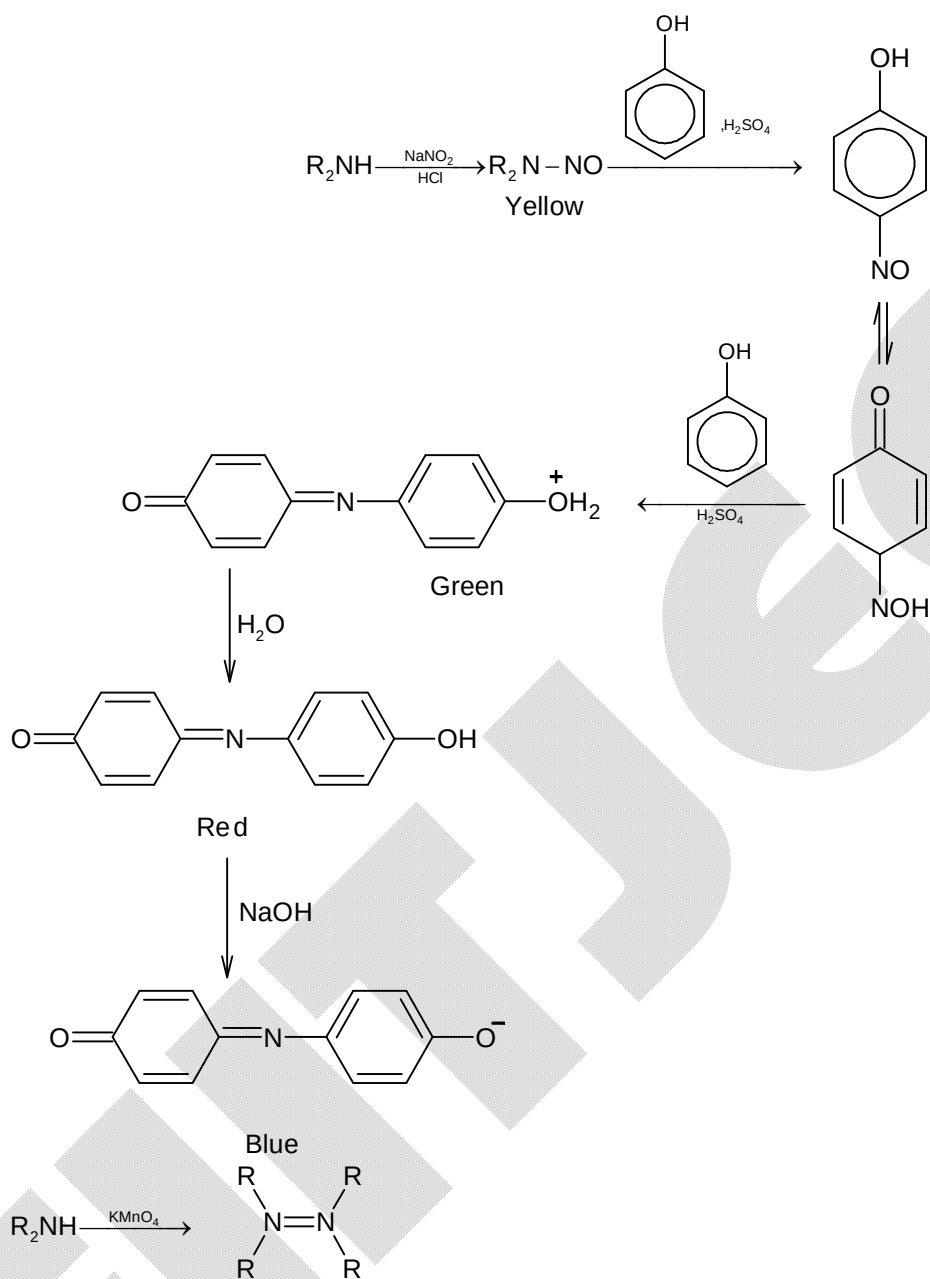
19. D



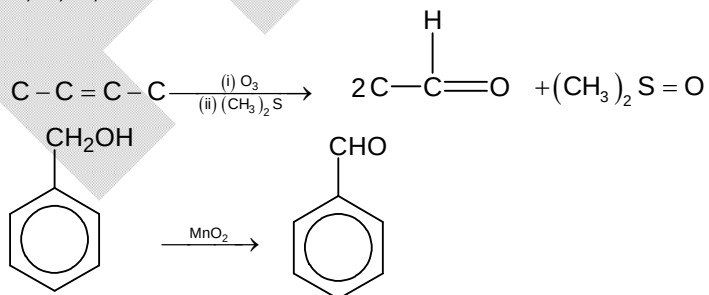
20. B
Sol.

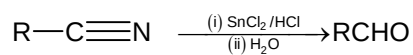


21. B
Sol. x is dialkyl amine

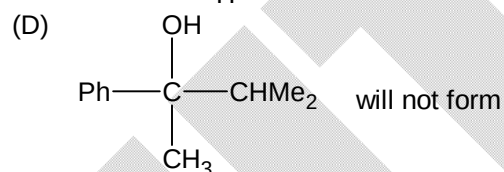
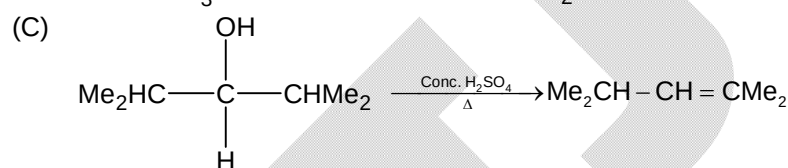
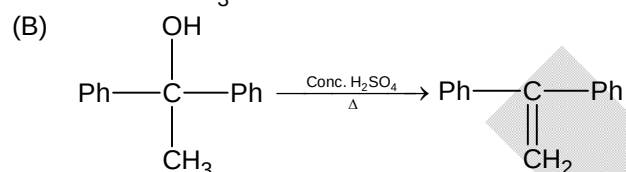
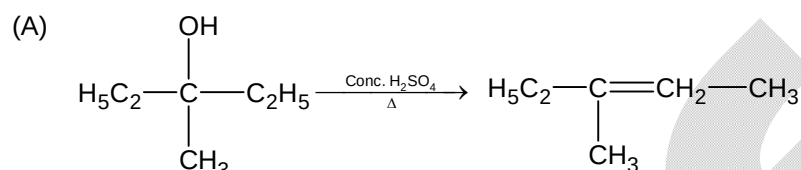
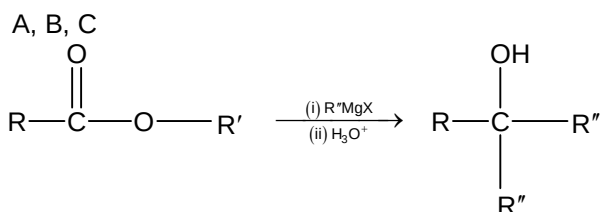


22. Sol. A, B, C, D

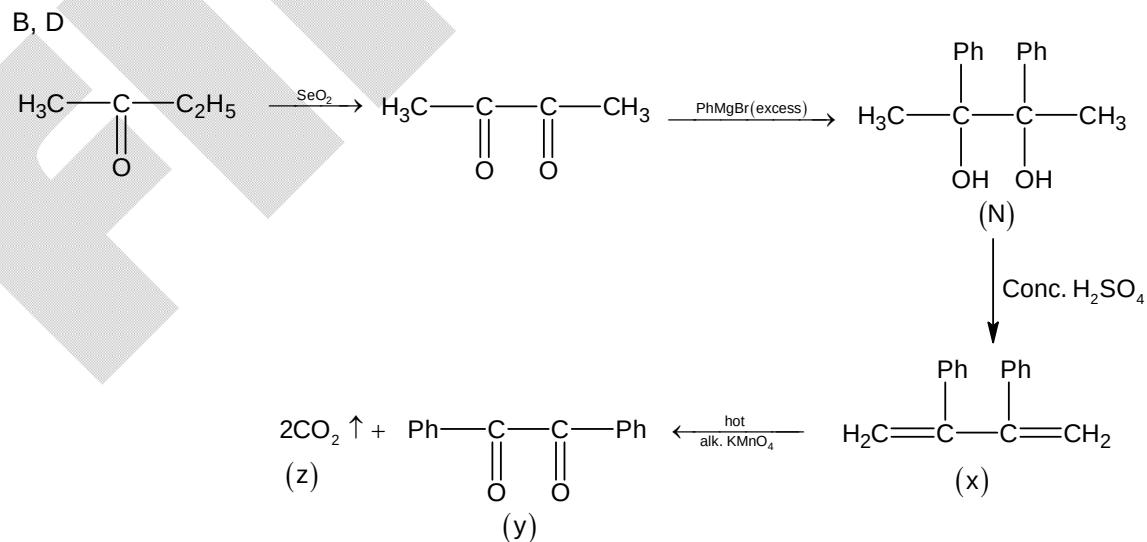


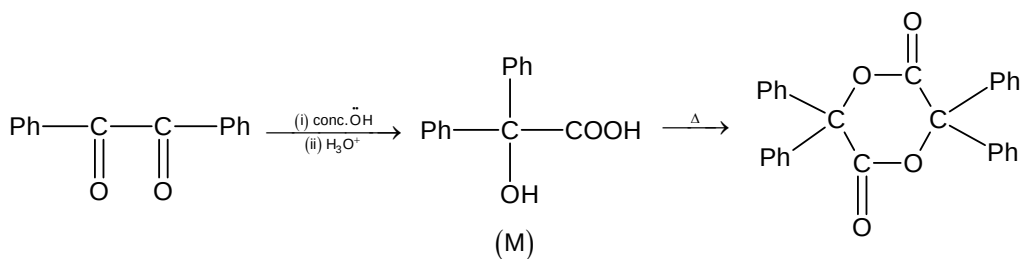


23.
Sol.



24.
Sol.

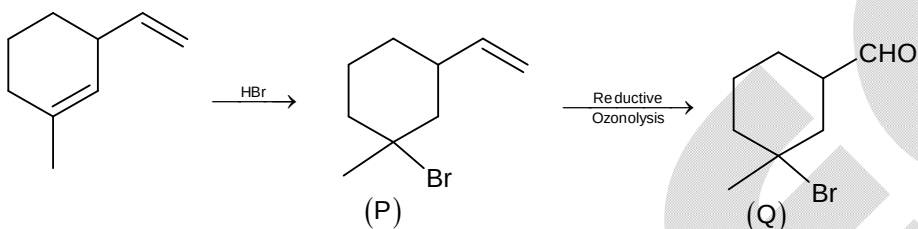




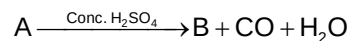
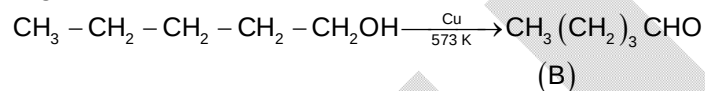
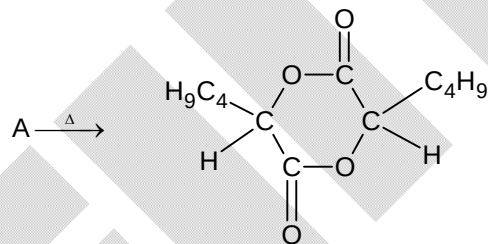
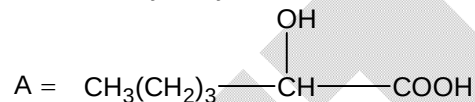
SECTION - B

25.
Sol.

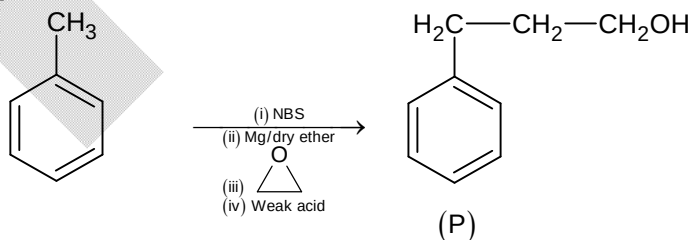
205

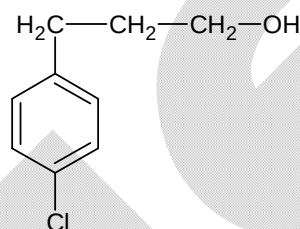
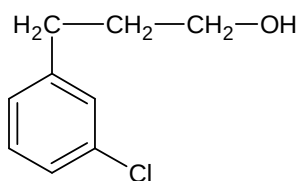
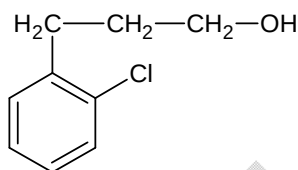
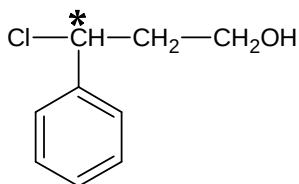
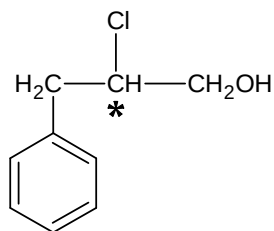
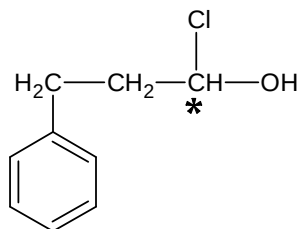
Mol. wt. of $\text{C}_8\text{H}_{13}\text{OBr} \Rightarrow 205$ 26.
Sol.

228

 $\therefore$ A is α -hydroxyl acid $\therefore$ Mol. wt. of $\text{B}(\text{C}_{12}\text{H}_{20}\text{O}_4) = 228$ 27.
Sol.

9

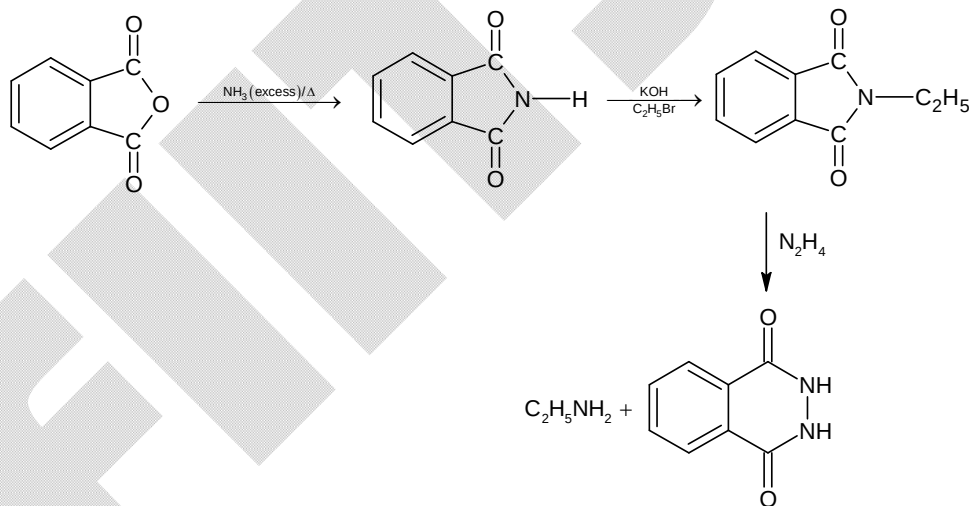




$$\text{Total} = 2 + 2 + 2 + 1 + 1 + 1 = 9$$

28.
Sol.

28



29.

6

Sol.

$$PI = \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{3.59 + 8.41}{2} = \frac{12}{2} = 6$$

30.

64

Sol.

$$\frac{(111 \times x) + (100 - x)20}{100} = 78.24$$

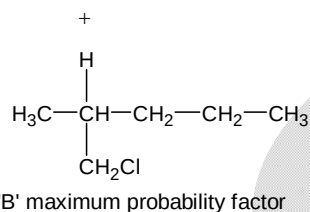
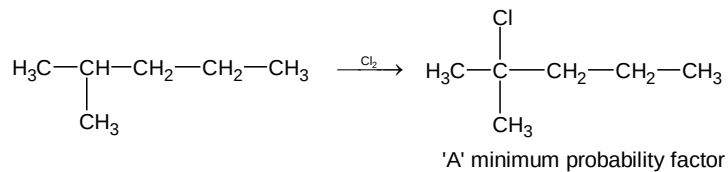
$$x = 64\%$$

SECTION – C

31. 17.12

32. 20.55

Solution for the Q. No. 31 and 32.

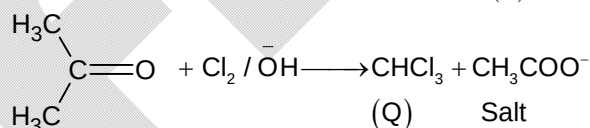
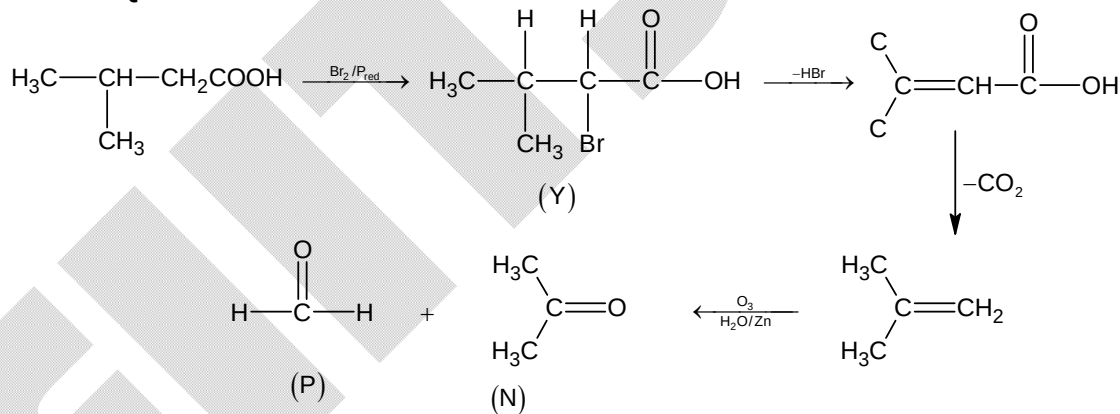


Product	reactivity factor × probability factor		=	number of P atoms	%
A	5	1		5	17.12%
B	1	6		6	20.55%

33. 180.90

34. 119.50

Solution for the Q. No. 33 and 34.



Molecular weight of Q = 12 + 1 + (35.5 × 3) = 119.5

Molecular weight of Y = (12 × 5) + (9 × 1) + 32 + 79.9 = 180.9

Mathematics

PART – III

SECTION – A

35.

A

 Sol. Equation of tangent at $P(8t^2, 16t)$ is $yt = x + 8t^2$... (1)

 $Q : (8 \cos \theta, 8 \sin \theta)$ which satisfies equation (1)

$$8 \sin \theta t = 8 \cos \theta + 8t^2$$

Equation of chord of contact AB is

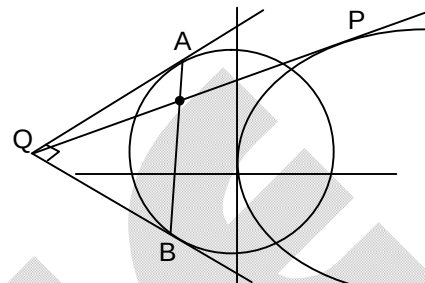
$$x \cos \theta + y \sin \theta = 4$$

$$x \cos \theta + y \left(\frac{\cos \theta + t^2}{t} \right) - 4 = 0$$

$$\cos \theta \left(x + \frac{y}{t} \right) + yt - 4 = 0$$

$$\Rightarrow y = \frac{4}{t}, x = -\frac{4}{t^2} \text{ point satisfies equation (1)}$$

$$4 = -\frac{4}{t^2} + 8t^2 \Rightarrow 8t^4 - 4t^2 - 4 = 0 \Rightarrow 2t^4 - t^2 - 1 = 0 ; t^2 = 1 \therefore t = \pm 1$$



36.

B

$$\text{Sol. } t_r = \cot^{-1} \frac{25r^2 - 15r + 21}{25} = \tan^{-1} \frac{1}{r^2 - \frac{3}{5}r + \frac{21}{25}} = \tan^{-1} \left(r + \frac{1}{5} \right) - \tan^{-1} \left(r - \frac{4}{5} \right)$$

$$\therefore S_n = \tan^{-1} \left(n + \frac{1}{5} \right) - \tan^{-1} \frac{1}{5} = \tan^{-1} \frac{25n}{5n + 26} \text{ hence, } x = \tan^{-1} 5$$

37.

C

 Sol. Equation of chord bisected at $P\left(\frac{5}{2}, \alpha\right)$ is $T = S_1$

$$\therefore 8x + \alpha y = \alpha^2 + 20 \text{ where } \frac{\alpha^2 + 20}{80} < 1$$

$$m(\text{slope}) = -\frac{8}{\alpha} \text{ and } |\alpha| \leq 7$$

$$\therefore m \in \mathbb{I} \therefore \alpha = \pm 1, \pm 2, \pm 4$$

38.

C

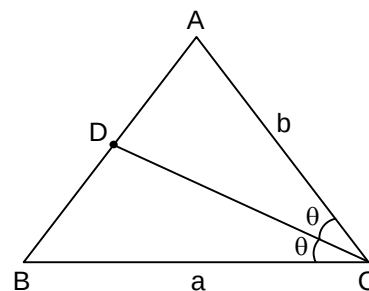
$$\text{Sol. } a = b + \frac{b^2}{a+b}$$

$$\Rightarrow a^2 + ab = ab + 2b^2$$

$$\Rightarrow a = \sqrt{2}b$$

$$\text{Hence, } \angle A = \frac{\pi}{2}$$

$$\therefore \angle B = \angle C = \frac{\pi}{4}$$



39. B, C

Sol. $t_n = t_1^{3^{n-1}}$; n is odd
 $= -\frac{1}{t_1^{3^{n-1}}}$; n is even

40. A, B, C, D

Sol. $a(e-1) = 5 \Rightarrow a = 20$
 $b^2 = 400\left(\frac{25}{16} - 1\right) = 225$

41. A, B, C

Sol. $\cos^{-1} \cos\left(4 + \frac{8}{x^2 + 4}\right) \leq 2\pi - 5 \Rightarrow 2\pi - \left(\frac{4x^2 + 24}{x^2 + 4}\right) \leq 2\pi - 5$
 $\Rightarrow 4x^2 + 24 \geq 5x^2 + 20 \Rightarrow x \in [-2, 2]$
 $9x + \alpha^2 y = 16$
 $x = 0, y = \frac{16}{\alpha^2} \Rightarrow \alpha^2 = 1, 4, 16$
 $x = 1, y = \frac{7}{\alpha^2} \Rightarrow \alpha^2 = 1$
 $x = 2, y = -\frac{2}{\alpha^2} \Rightarrow \alpha^2 = 1$
 $x = -1, y = \frac{25}{\alpha^2} \Rightarrow \alpha^2 = 1, 25$
 $x = -2, y = \frac{34}{\alpha^2} \Rightarrow \alpha^2 = 1$

SECTION - B

42. 2

Sol. $h > 2$ and $\frac{h^2}{50} + \frac{k^2}{4} < 1$ & $(h-2)^3 < \frac{27k^2}{4}$

If $h = 3, |k| \leq 1$ & $|k| \geq 1$ $h = 4, |k| \leq 1$ & $|k| \geq 2$ not possible $h = 5, |k| \leq 1$ & $|k| > 2$ not possible $\therefore h = 3, k = \pm 1$

43. 6

Sol. $4R \cos 36^\circ = 4(\sqrt{5} + 1) \Rightarrow R = 4$

$$\Delta DEF = \frac{1}{2} R^2 \sin 2A \sin 2B \sin 2C = 8 \sin 72^\circ \sin^2 36^\circ$$

$$k^2 = \frac{5}{2}(5 - \sqrt{5})$$

44. 1012

Sol. $(\sqrt{3} + 1)\sin x + (\sqrt{3} - 1)\cos x = 2 \Rightarrow \cos\left(x - \frac{5\pi}{12}\right) = \cos \frac{\pi}{4} \Rightarrow x - \frac{5\pi}{12} = 2n\pi \pm \frac{\pi}{4}$

$$\therefore x = 2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{\pi}{6}; n \in \mathbb{I}$$

45. 676

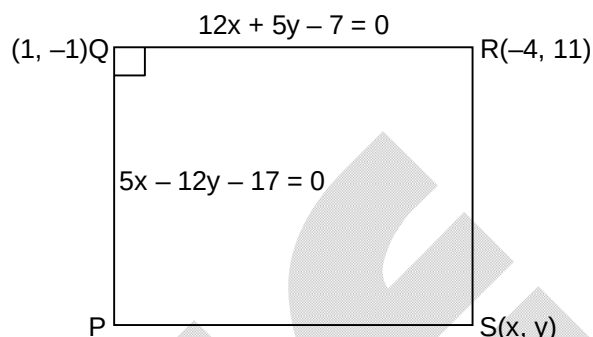
Sol. Foot of perpendicular from R(-4, 11) on the line $5x - 12y - 17 = 0$ is (1, -1)

Also, (1, -1) is point of intersection of the given lines

$$\therefore Q(1, -1)$$

Hence, PQRS is a square therefore locus of S is

$$(x+4)^2 + (y-11)^2 = \frac{(5x-12y-17)^2}{169}$$



46. 4

$$\text{Sol. } \frac{\tan\left(x + \frac{\pi}{9}\right)}{\tan x} = \tan\left(x + \frac{\pi}{18}\right) \tan\left(x - \frac{\pi}{18}\right)$$

$$\Rightarrow \frac{\sin\left(2x + \frac{\pi}{9}\right)}{\sin \frac{\pi}{9}} = \frac{-\cos \frac{\pi}{9}}{\cos 2x} \Rightarrow \sin\left(4x + \frac{\pi}{9}\right) + \sin \frac{\pi}{9} = -\sin \frac{2\pi}{9}$$

$$\Rightarrow \sin\left(4x + \frac{\pi}{9}\right) = -\cos \frac{\pi}{18} = -\sin \frac{4\pi}{9} \Rightarrow 4x + \frac{\pi}{9} = \frac{13\pi}{9}, \frac{14\pi}{9}$$

$$\therefore x = \frac{\pi}{3}, \frac{13\pi}{36}$$

47. 6

Sol. Equation of the circle $S = 0$ is $(x+14)^2 + (y+7)^2 - 15^2 = 0$

Point on the circle is $(-14 + 15 \cos \theta, -7 + 15 \sin \theta)$

$$\therefore (15 \cos \theta - 9)^2 + (15 \sin \theta - 12)^2 = 90$$

$$\Rightarrow 4 \sin \theta + 3 \cos \theta = 4 \Rightarrow \sin \theta = \frac{7}{25}, \cos \theta = \frac{24}{25} \text{ or } \sin \theta = 1, \cos \theta = 0$$

SECTION - C

48. 8.00

49. 26.00

Sol. (Q.48.-49)

C : $(3 \sec \theta, 2 \tan \theta)$

$$\frac{\sum x_i}{4} = \frac{3 \sec \theta}{2}; \frac{\sum y_i}{4} = \frac{2 \tan \theta}{2} \Rightarrow \sum x_i = 6 \sec \theta; \sum y_i = 4 \tan \theta$$

If (h, k) be the centroid of ΔPQR , then $3h = x_1 + x_2 + x_3, 3k = y_1 + y_2 + y_3$

$$3h + x_4 = 6 \sec \theta, 3k + y_4 = 4 \tan \theta \Rightarrow x_4 = (6 \sec \theta - 3h), y_4 = (4 \tan \theta - 3k)$$

$$\text{Hence, } (h - 2 \sec \theta) \left(k - \frac{4}{3} \tan \theta \right) = \frac{72}{9} = 8 \therefore \text{Latus rectum} = 8$$

(i) Focus $x = 2 \sec \theta + 4$, $y = \frac{4}{3} \tan \theta + 4$ $\therefore$ Locus of focus is $\frac{(x-4)^2}{4} - (y-4)^2 \times \frac{9}{16} = 1$

$$\therefore e^2 = 1 + \frac{16}{9 \times 4} = \frac{13}{9}$$

50. 4.00

51. 5.00

Sol. (Q.50.-51)

$$a = 20\pi - 60, b = 50 - 16\pi, c = 3\pi - 10$$

$$\cos^{-1} \cos x \geq |x - \pi| \Rightarrow k = 4$$

$$\pi - \cos^{-1} \left(4 \left(\frac{k}{10} \right)^3 - \frac{3k}{10} \right) = \pi - \left(2\pi - 3 \cos^{-1} \frac{k}{10} \right) = 3 \cos^{-1} \frac{k}{10} - \pi$$

$$\alpha = 3, \beta = -\pi \therefore \alpha - \frac{2\beta}{\pi} = 3 + 2 = 5$$